



NoraSol

PRODUCT CATALOGUE

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Vision

***“ENHANCING ECONOMIC GROWTH IN PAKISTAN THROUGH
INCREASED ENERGY ACCESSIBILITY AND AFFORDABILITY”***



First in Pakistan to carry out in-house R&D on:

Solar Inverters

Battery Management Systems (BMS)

Smart Energy Solutions

Shift from import-dependence
→ local innovation

Reduced cost of ownership

Increased accessibility for low-
income groups

NORASOL TEAM

- Subsidiary of CARE Pvt. Ltd., a leading R&D organization working on defense technologies since 2003.
- NoraSol team comprises 15 highly trained R&D and mechanical design experts.
- Specialized in in-house design and development of:
 - Inverter Hardware
 - Battery Management System (BMS)
 - Related software solutions

PRODUCT LINEUP

Nora-3



3 kW Hybrid inverter featuring smart storage (3 kWh) Battery Energy Storage System (BESS) all combined into a single unit, complete with built-in Wi-Fi monitoring.

NoraStore-16



51.2V 314Ah long life Lithium Iron Phosphate Battery

NoraStore-5



51.2V 100Ah long life Lithium Iron Phosphate Battery

NoraStore-12



51.2V 230Ah long life Lithium Iron Phosphate Battery

PRODUCT LINEUP

Our Product Features:

We offer a range of products, including hybrid solar inverters and lithium iron phosphate battery packs, featuring the following key attributes:

- **Smart Power Switching:** Enjoy continuous electricity sourced from solar, the grid, or battery.
- **Solar Self-Consumption Enabled:** Our inverters ensure zero grid export while meeting home energy demands, eliminating the need for net metering or green meters.
- **Energy Storage:** Benefit from long-lasting lithium batteries equipped with an intelligent battery management system (BMS).
- **High Efficiency:** Maximize solar power usage while reducing grid expenses.
- **Smart Monitoring:** Utilize Wi-Fi and an app to monitor your energy consumption at any time.
- **Designed and Made in Pakistan:** Indigenously crafted to withstand heat, dust, and power outages.



NORA-3

Inverter Specs



Output Waveform	Pure Sine Wave – clean and stable power
MPPT Range	120V – 430V DC
MPPT Efficiency	≥ 98% – maximum solar energy harvesting
Battery Voltage Range	200V – optimized for efficient storage
Monitoring	Built-in Wi-Fi for cloud monitoring through dedicated Mobile Application
Input Sources	Solar / Grid / Battery (Hybrid)

NORA-3

Battery Specs

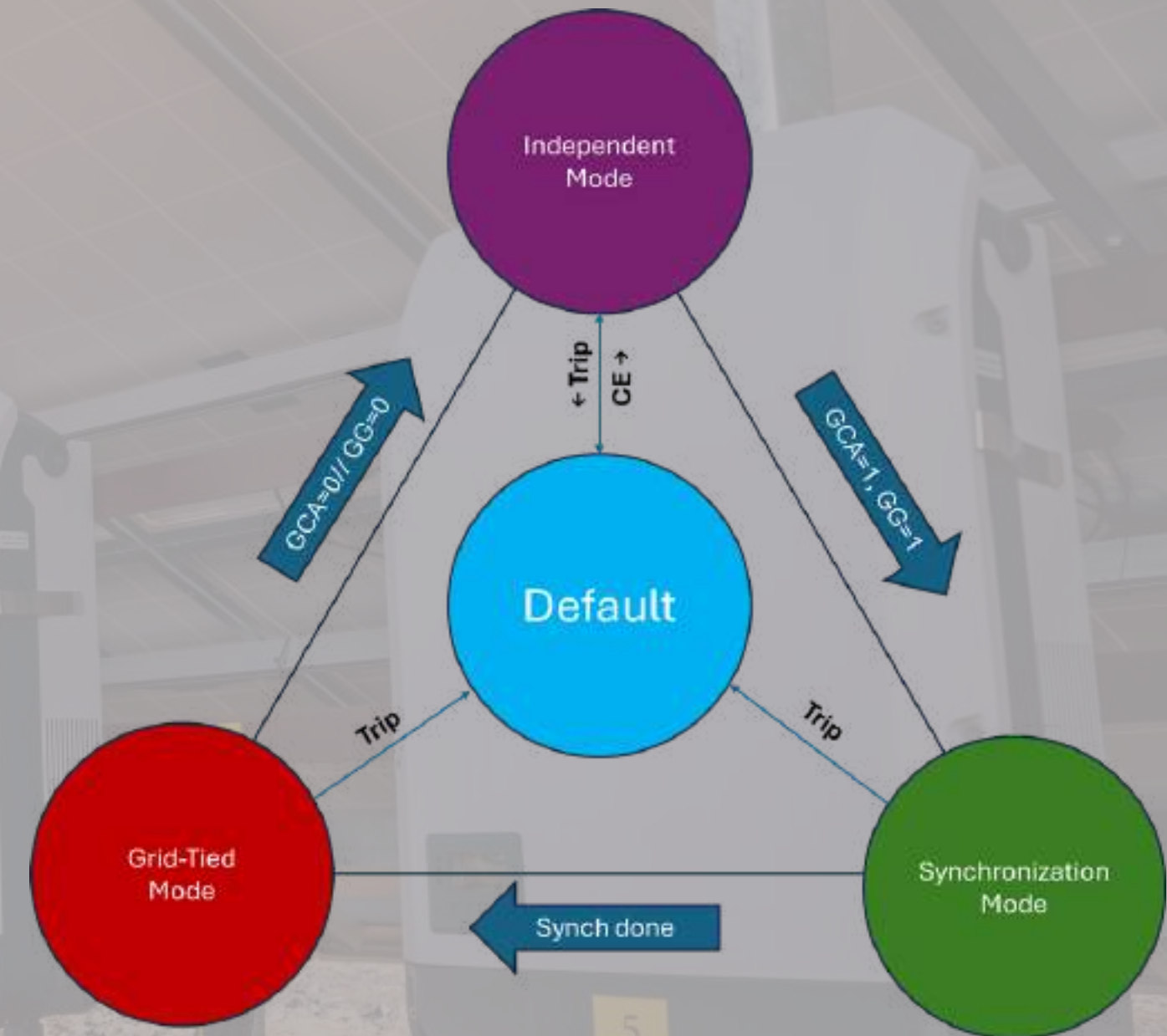


Type	Lithium Iron Phosphate (LiFePO ₄)
Nominal Voltage	204V
Capacity	15Ah (3.0 kWh)
Cycle Life	6,000 cycles
Design	Wall-mount, modular & expandable
Protection	BMS with Active Balancing, Over/Under Voltage, Temperature & Short Circuit

NORA-3

Hybrid inverter Operating Modes

- **Grid-Tied:** Uses grid power when necessary
- **Off-Grid:** Works with solar panels and batteries in areas without grid access.
- **Hybrid:** Combines solar, battery, and grid energy, adapting to resources.
- **Backup:** Provides power during outages using battery backup.
- **Self-Consumption:** Prioritizes solar for on-site use, reducing grid reliance (With Zero Export Capability).
- **Time-of-Use:** Optimizes use based on pricing for cost efficiency.
- **Forced:** Allows manual selection of one energy source.



NoraStore-16

Battery Technical Specifications



Battery Type	LiFePO ₄
Nominal Voltage	51.2V
Capacity	16kWh (314Ah)
Cycle Life	>6000 cycles
Communication/Monitoring	CAN / RS485 + Built-in Wi-Fi
Protection	BMS with Active Balancing, Over/Under Voltage, Temperature & Short Circuit

Battery Type	LiFePO ₄
Design	Modular, rack-mount, expandable
Nominal Voltage	51.2V
Capacity	Expandable (5-30kWh) Upto 6 Units can be Stacked
Cycle Life	>6000 cycles
Communication / Monitoring	CAN / RS232, RS485 + Built-in Wi-Fi
Protection	BMS with Active Balancing, Over/Under Voltage, Temperature & Short Circuit

NoraStore-5

Battery Technical Specifications



NoraStore-12

Battery Technical Specifications

Battery Type	LiFePO ₄
Nominal Voltage	51.2V
Capacity	230Ah
Cycle Life	>6000 cycles
Communication	CAN / RS485 + Built-in Wi-Fi
Protection	BMS with Active Balancing, Over/Under Voltage, Temperature & Short Circuit



Final Remarks

Reliable power during outages

Smart monitoring & safety for users

Lower energy costs through solar efficiency

Reduction in import dependency and open
doors for exports

Together, we can power Pakistan's future sustainably
one home, one business, one community at a time.

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